

E-Commerce Sales & Customer Behavior Analysis

Executive Summary

This project analyzes 2.5 years of e-commerce transactional data to uncover the factors driving revenue performance, customer purchasing behavior, marketing-channel effectiveness, and operational inefficiencies. Using Python, Pandas, DuckDB, SQL, Seaborn, and Matplotlib, the analysis combines statistical exploration, visualization, and business reasoning to transform raw transactional records into actionable insights.

The study reveals that revenue fluctuations are primarily driven by changes in customer spending behavior rather than variations in demand volume. Marketing-channel analysis highlights significant differences in customer quality across acquisition sources, while operational analysis identifies elevated cancellation tendencies among specific channels. Revenue-concentration analysis further demonstrates a relatively diversified revenue structure, reducing dependence on a small number of high-value transactions.

Business Objective

The objective of this analysis was to answer the following business questions:

- What factors drive revenue growth and revenue volatility?
- How do customers behave across different acquisition channels?
- Which referral sources generate the highest-quality customers?
- What causes high cancellation rates within the order pipeline?
- How are pricing, quantity, and cart behavior related to revenue generation?
- Is revenue concentrated among a small number of transactions or distributed across the customer base?

Methodology

The analysis was conducted in four stages:

1. Data Validation & Quality Checks

- Verified data types
- Inspected missing values
- Checked duplicate records
- Validated pricing consistency
- Investigated outliers

2. Exploratory Data Analysis

- Distribution analysis
- Frequency analysis
- Customer behavior analysis
- Revenue analysis

3. Relationship Analysis

- Correlation analysis
- Scatterplot analysis
- Referral-source comparisons
- Operational status comparisons

4. Time-Series Analysis

- Monthly revenue trends
- Rolling averages
- Order-volume analysis
- Average Order Value (AOV) analysis

Key Findings

1. Revenue Volatility Is Driven by Customer Spending Behavior

Monthly revenue exhibited significant fluctuations throughout the observation period.

However, comparison of:

- Monthly Revenue
- Monthly Order Count
- Monthly Average Order Value (AOV)

revealed that order volume remained relatively stable while AOV fluctuated substantially.

Insight

Revenue volatility is primarily driven by variations in customer spending behavior rather than significant changes in customer demand.

Business Impact

Improving customer spending through pricing strategies, premium-product promotion, and upselling initiatives may produce greater revenue gains than focusing exclusively on customer acquisition.

2. Instagram Generated the Strongest Overall Channel Performance

Referral-source analysis showed meaningful differences in customer behavior across acquisition channels.

Key Observations

- Instagram generated the highest total revenue.
- Instagram maintained the lowest cancellation rate (~16%).
- Google and Email channels exhibited the highest cancellation rates (~24%).

Insight

Instagram appears to attract higher-quality customers who are more likely to complete purchases successfully.

Business Impact

Marketing investments may produce stronger returns when allocated toward acquisition channels demonstrating both strong revenue generation and lower operational leakage.

3. Facebook Customers Demonstrated Higher Spending Per Transaction

Although Facebook generated fewer total orders than Instagram, it recorded the highest Average Order Value (AOV).

Insight

Facebook customers exhibit stronger per-order spending behavior.

Business Impact

Facebook may be particularly effective for promoting premium products or higher-value offerings.

4. Product Pricing Is the Strongest Revenue Driver

Correlation analysis revealed:

Relationship	Correlation
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UnitPrice ↔ TotalPrice	0.72
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Quantity ↔ TotalPrice	0.62
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ItemsInCart ↔ TotalPrice	0.39
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Insight

Revenue generation is influenced most strongly by product pricing, followed by purchase quantity.

Business Impact

Pricing strategy and product positioning have a greater impact on revenue than increasing cart size alone.

5. High-Value Orders Were Legitimate Transactions

Outlier investigation identified several unusually large transactions.

Detailed inspection revealed that these orders were associated with:

- Higher purchase quantities
- Larger cart sizes
- Consistent pricing behavior

Insight

High-value transactions represent legitimate customer purchasing behavior rather than data-quality issues.

Business Impact

Premium and bulk-purchase customers contribute meaningful revenue and should not be excluded during analysis.

6. Revenue Is Not Excessively Concentrated

Pareto analysis was performed to evaluate revenue concentration.

Key Observation

Approximately 50% of orders contributed to nearly 80% of cumulative revenue.

Insight

Revenue generation is relatively diversified across the customer base.

Business Impact

The business is less vulnerable to losing a small number of high-value customers compared to organizations with highly concentrated revenue structures.

Operational Analysis

Order-status analysis revealed an unusually high proportion of cancelled orders.

Further investigation showed that cancellation behavior varied by acquisition channel rather than being concentrated within a single operational process.

Highest Cancellation Channels

- Google
- Email

Lowest Cancellation Channel

- Instagram

Insight

Customer acquisition quality plays a significant role in operational outcomes and order completion rates.

Time-Series Insights

Revenue trends showed:

- Significant month-to-month fluctuations
- Periodic revenue spikes
- Limited evidence of sustained long-term growth

Rolling-average analysis revealed:

- Temporary revenue surges were not consistently maintained.
- Revenue momentum weakened during several periods before recovering.
- Customer demand remained relatively stable despite fluctuations in revenue.

Insight

Revenue instability appears linked to spending behavior rather than major shifts in customer demand.

Recommendations

Marketing Strategy

- Increase investment in Instagram campaigns.
- Review targeting effectiveness of Google and Email campaigns.
- Optimize acquisition channels based on customer-quality metrics rather than order volume alone.

Revenue Growth

- Focus on premium-product promotion.
- Implement upselling and cross-selling initiatives.
- Encourage larger transaction values through bundling strategies.

Operational Improvement

- Monitor cancellation rates by referral source.
- Investigate causes of purchase abandonment among Google and Email traffic.

Customer Strategy

- Identify and retain high-value customers.
 - Develop targeted campaigns for customers exhibiting premium purchasing behavior.
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Conclusion

This analysis demonstrates how transactional data can be leveraged to uncover meaningful business insights regarding customer behavior, revenue generation, marketing effectiveness, and operational performance.

The findings indicate that revenue volatility is driven primarily by fluctuations in customer spending rather than demand volume, while acquisition-channel quality significantly influences both revenue performance and cancellation behavior. Product pricing and purchase quantity emerged as the strongest drivers of transaction value, and revenue was found to be moderately distributed across the customer base rather than concentrated among a small subset of transactions.

By combining SQL-based analysis, statistical reasoning, and visualization techniques, the project successfully transformed raw transactional data into actionable business intelligence capable of supporting strategic decision-making.